Project Title

ESP32-Based Smart Health Monitoring Belt with ECG, SpO_2, Temperature, OLED Display, ThingSpeak Cloud, and Machine Learning

Abstract

The proposed project is an **ESP32-based Smart Health Monitoring Belt** designed to provide continuous, portable, and intelligent monitoring of important physiological parameters. The system integrates an **ECG sensor, SpO_2 sensor, and body-temperature sensor** with an ESP32 microcontroller to collect real-time health data from the user. The measured parameters are processed by the ESP32 and displayed locally on an **OLED display**, allowing the user to observe vital signs without requiring a smartphone or computer. At the same time, the collected data is transmitted through Wi-Fi to the **ThingSpeak cloud platform**, where it can be stored, visualized, and monitored remotely. A dedicated web application is incorporated to maintain the user's profile, including additional parameters such as **age, gender, height, weight, existing health information, and other relevant personal parameters**. These profile parameters are combined with the sensor measurements to create a more comprehensive input for a **machine-learning model**. The ML system analyzes patterns in ECG, SpO_2, temperature, and user-profile information to classify or predict the user's overall health status, such as **Normal, Attention Required, or Critical/Risk**. The system is intended as a low-cost prototype for continuous health monitoring and early indication of abnormal conditions rather than as a replacement for certified medical equipment. The combination of wearable sensing, cloud connectivity, web-based profile management, and machine learning provides an intelligent framework for remote and personalized health-status monitoring.

Introduction

Continuous monitoring of physiological parameters is becoming increasingly important because many health abnormalities can develop gradually and may not be recognized immediately by the individual. Conventional health monitoring generally requires dedicated medical instruments and periodic visits to healthcare facilities, making continuous observation inconvenient for many users. A wearable smart belt can provide a more convenient approach by integrating multiple physiological sensors into a compact wearable platform. In this project, an **ESP32** acts as the central controller because it provides sufficient processing capability, multiple interfaces for sensors, and built-in Wi-Fi connectivity for cloud communication. An **ECG sensor** is used to acquire electrical activity of the heart, while an **SpO_2 sensor** measures blood oxygen saturation and pulse-related information. A temperature sensor measures body temperature, providing another important indicator of the user's physiological condition. These measurements are continuously processed and presented on an OLED display attached to the belt, enabling immediate access to the measured values.

The project further extends conventional wearable monitoring by incorporating **cloud computing and machine learning**. Instead of limiting the system to displaying individual sensor readings, the ESP32 can transmit measurements to ThingSpeak through Wi-Fi, enabling historical data storage and visualization. The web application provides a user profile containing additional parameters that can influence health-status interpretation. For example,

the same physiological reading may have different significance depending on factors such as age, body characteristics, and other profile information. Therefore, the proposed ML model can combine the real-time sensor parameters with selected profile parameters to produce a more personalized health-status prediction. The system can continuously evaluate incoming measurements and provide a simplified status indication through the OLED and web interface. This architecture demonstrates how **IoT, wearable electronics, cloud platforms, web applications, and machine learning** can be integrated into a single health-monitoring solution.

Scope of the Project

The scope of this project covers the development of a wearable health-monitoring belt capable of acquiring and processing multiple physiological parameters in real time. The belt primarily monitors **ECG, SpO_2, and body temperature**, with the ESP32 serving as the main processing and communication controller. The OLED display provides immediate visualization of the measured parameters, while Wi-Fi connectivity allows the system to upload data to ThingSpeak for remote monitoring and historical analysis. The project can be extended with a web application through which users can create and maintain their health profiles and view both current and historical measurements. The ML component can use the combination of sensor readings and selected profile information to estimate an overall health-status category and identify potentially abnormal patterns.

The project also provides scope for future expansion into more advanced wearable healthcare systems. Additional sensors such as **blood-pressure sensors, motion sensors, respiratory-rate sensors, or fall-detection sensors** can be integrated depending on the application requirements. The ML model can be improved using larger and clinically validated datasets, allowing better personalization and potentially supporting prediction of specific risk categories. Cloud-based dashboards can also be enhanced with alerts, trend analysis, notifications, and caregiver access. However, the proposed prototype should be considered an **assistive monitoring and research system**, and its ML predictions should not be treated as a medical diagnosis without appropriate clinical validation and regulatory approval.

Problem Statement

Traditional health monitoring systems often depend on separate medical devices for measuring different physiological parameters. Users may need to manually record ECG-related information, oxygen saturation, and temperature readings, making continuous monitoring difficult and inconvenient. Many basic wearable systems provide only real-time sensor values and do not maintain sufficient historical information for analyzing health trends. Furthermore, systems that simply compare sensor readings against fixed threshold values may not adequately account for individual differences between users. A health value that is acceptable for one person may require different interpretation depending on the person's age, body characteristics, and other relevant profile information. There is therefore a need for an integrated wearable system that can collect multiple physiological parameters, display them locally, store them remotely, and combine them with user-specific information for intelligent health-status estimation.

The proposed system addresses this problem by developing a smart belt based on ESP32 that continuously acquires ECG, SpO_2, and temperature measurements. The system provides immediate feedback through an OLED display and uploads measurements to ThingSpeak for

cloud-based monitoring. A web application stores the user's profile and allows additional parameters to be incorporated into the ML analysis. Instead of relying only on individual threshold values, the machine-learning component can learn relationships between physiological measurements and user-specific parameters from an appropriately prepared dataset. The resulting system can provide an easy-to-understand health-status classification and support early identification of measurements that may require further attention. The system is intended to support monitoring and awareness, not to independently diagnose disease.

Existing System

Existing health-monitoring solutions generally fall into three major categories: conventional medical instruments, standalone wearable devices, and IoT-enabled monitoring systems. Conventional systems such as clinical ECG equipment and pulse oximeters can provide relatively accurate measurements when properly used, but they are generally designed for clinical or point-of-care environments and may not be convenient for continuous everyday monitoring. Standalone wearable devices have improved portability and provide users with easier access to physiological measurements. However, many low-cost devices primarily display individual readings and may provide limited cloud storage, personalization, or intelligent analysis. In some systems, the user must manually interpret the values or depend on predefined threshold limits to determine whether a measurement is normal or abnormal.

IoT-based health-monitoring systems have addressed some of these limitations by connecting wearable sensors to cloud platforms. Such systems can transmit sensor data to remote servers and allow doctors, caregivers, or users to access measurements through dashboards. However, many implementations concentrate on data collection and visualization rather than combining multiple physiological parameters with individual user characteristics. Systems based solely on threshold-based decisions can also generate unnecessary warnings because physiological measurements can naturally vary between individuals and under different conditions. Therefore, although existing systems provide useful monitoring capabilities, there remains an opportunity to develop a more integrated platform combining **wearable multi-parameter sensing, local OLED visualization, cloud monitoring, user-profile information, and machine-learning-based health-status prediction**.

Proposed System

The proposed system introduces an intelligent wearable belt that integrates **ESP32, ECG sensor, SpO_2 sensor, temperature sensor, OLED display, ThingSpeak cloud, web application, and machine-learning technology** into a unified architecture. The sensors attached to the belt continuously collect physiological information from the user. The ECG sensor captures electrical cardiac activity, the SpO_2 sensor measures oxygen saturation and pulse-related information, and the temperature sensor measures body temperature. The ESP32 receives these sensor signals, performs initial processing and validation, and prepares the measurements for display and transmission. The OLED display provides real-time information such as ECG-related readings, SpO_2 percentage, temperature, and the current estimated health status. This local display makes the system useful even when the user does not have immediate access to the web application.

The ESP32 uses its built-in Wi-Fi capability to transmit the processed sensor readings to **ThingSpeak**, where the data can be stored and visualized as time-series information. This

enables the development of graphs and historical health trends rather than relying only on instantaneous readings. A separate web application maintains the user's profile and can include parameters such as age, gender, height, weight, and other non-sensitive information relevant to the selected ML model. The web application can retrieve or receive the sensor data and combine it with the user's profile information before sending the feature set to the machine-learning model. The ML model can be trained using an appropriately labeled health dataset containing physiological measurements and corresponding health-status classes. During operation, the model analyzes the incoming measurements and produces an estimated status such as **Normal, Attention Required, or Critical/Risk**, depending on the classes defined during training.

The overall system therefore follows a continuous monitoring cycle: **sensor data acquisition → ESP32 processing → OLED display → ThingSpeak cloud storage → web application → profile-data integration → ML analysis → health-status prediction**. If abnormal patterns are detected, the system can provide a warning through the OLED display and web interface. Historical cloud data can also be used to observe changes over time and improve the ML model in future versions. The proposed architecture is particularly suitable as an academic IoT and ML project because it demonstrates the integration of embedded systems, biomedical sensing, wireless communication, cloud computing, web development, data analytics, and artificial intelligence in one practical application.

Literature Survey

1. “Wearable ECG Device and Machine Learning for Heart Monitoring”

Authors: Zhadyra Alimbayeva, Chingiz Alimbayev, Kassymbek Ozhikenov, Nurlan Bayanbay, Aiman Ozhikenova
Year: 2024

Abstract:
Alimbayeva et al. presented a wearable ECG monitoring system combined with machine-learning techniques for continuous cardiac monitoring. The research addresses the increasing demand for portable cardiovascular monitoring caused by the growing prevalence of cardiovascular diseases. The proposed approach focuses on acquiring ECG signals using wearable hardware and applying computational methods to analyze cardiac activity outside conventional clinical environments. Machine-learning techniques are incorporated to improve the interpretation of ECG signals and assist in identifying cardiac abnormalities. The study demonstrates how wearable ECG acquisition can be combined with intelligent signal-processing techniques to provide more accessible cardiac monitoring. Such systems can reduce the dependency on large clinical equipment and enable physiological information to be collected during normal daily activities. The work is particularly relevant to the proposed smart belt because it establishes the feasibility of combining wearable ECG sensing with machine learning for intelligent health assessment.

Pros:

- Combines wearable ECG sensing with machine learning.
- Suitable for continuous cardiac monitoring.
- Reduces dependence on conventional clinical equipment.

- Supports intelligent ECG interpretation.

Cons:

- Primarily focused on ECG rather than complete multi-parameter monitoring.
- ECG signals are susceptible to motion artifacts.
- Clinical validation is required before diagnostic use.

2. "A Comprehensive Health Assessment Approach Using Ensemble Deep Learning Model for Remote Patient Monitoring with IoT"

Authors: Gayathri R, Maheswari S, Sandeep Kumar Mathivanan, Basu Dev Shivahare, Radha Raman Chandan, Mohd Asif Shah et al.
Year: 2024

Abstract:
This research proposes an ensemble deep-learning approach for remote patient monitoring using Internet of Things technology. The study combines **Convolutional Neural Networks (CNN)** and **Long Short-Term Memory (LSTM)** networks to analyze health-related information collected through IoT-based monitoring systems. The CNN component is useful for extracting meaningful patterns from physiological data, while LSTM networks are suitable for identifying temporal relationships in continuously acquired health signals. The research demonstrates the importance of combining multiple deep-learning architectures for comprehensive health assessment rather than depending on a single physiological measurement. The IoT architecture enables remote collection and analysis of patient information, supporting continuous monitoring outside hospitals. The work provides an important foundation for the proposed smart belt because the proposed system can similarly combine multiple sensor measurements and use ML to classify overall health status.

Pros:

- Combines CNN and LSTM for improved health-data analysis.
- Suitable for remote patient monitoring.
- Supports analysis of temporal physiological patterns.
- Demonstrates the potential of IoT and deep learning together.

Cons:

- Deep-learning models require substantial training data.
- Computational requirements can be higher than conventional ML.
- Cloud-based processing can introduce latency and privacy concerns.

3. "Enhanced Real-Time Analysis and Anomaly Detection in Smart Health Monitoring Systems through Integration of Deep Learning Algorithm with IoT-Edge Computing"

Authors: Authors associated with the IEEE Conference publication
Year: 2024

Abstract:
This study introduces an intelligent healthcare monitoring framework called **HealthEdge**, combining IoT, edge computing, multiple physiological sensors, and deep-learning-based anomaly detection. The system considers a diverse collection of physiological parameters including respiratory rate, pulse rate, SpO_2, galvanic skin response, body temperature, and other health-related signals. Instead of sending all raw information directly to a remote cloud server, edge computing is used to support real-time analysis closer to the data source. This approach can reduce communication latency and allow potentially abnormal health conditions to be detected more quickly. The multi-sensor architecture is particularly relevant to wearable healthcare because health conditions are often better characterized using multiple physiological parameters rather than a single measurement. The study provides a strong basis for the proposed project, where ESP32 can collect ECG, SpO_2, and temperature data and cloud/edge ML techniques can be used for health-status prediction.

Pros:

- Supports real-time anomaly detection.
- Uses multiple physiological parameters.
- Edge computing can reduce response latency.
- Suitable for intelligent IoT healthcare systems.

Cons:

- Multi-sensor systems increase hardware complexity.
- Edge AI requires optimization for resource-constrained devices.
- Sensor noise can affect prediction performance.

4. "AI-Driven IoT Healthcare System for Real-Time ECG Analysis and Comprehensive Patient Monitoring"

Authors: Mahdi Seddiki, Mounir Bouhedda, Abdelhafid Tobb al, Samia Rebouh, Yehya Aniba
Year: 2024

Abstract:
Seddiki et al. developed an IoT healthcare architecture focused on real-time ECG analysis and comprehensive patient monitoring. The system uses deep-learning models trained using the MIT-BIH ECG dataset, with one model dedicated to R-R peak detection and another model designed to classify ECG signals into multiple disease categories. In addition to ECG analysis, the proposed IoT architecture collects physiological parameters such as **ECG, SpO_2, and temperature** using an ESP32 and Raspberry Pi-based platform. The collected information can be transmitted for remote analysis, thereby combining biomedical sensing with IoT communication and artificial intelligence. The study reported high classification performance for its ECG models and demonstrated the feasibility of integrating physiological sensors with intelligent analysis. This paper is particularly important for the proposed smart belt because it closely matches the combination of ECG, SpO_2, temperature, ESP32-based acquisition, and ML-based health analysis.

Pros:

- Directly combines ECG, SpO_2, and temperature.
- Uses ESP32-based physiological data acquisition.
- Applies deep learning to ECG classification.
- Supports real-time remote healthcare monitoring.

Cons:

- ECG classification depends strongly on the quality of training datasets.
- Deep-learning models may require considerable computational resources.
- Prototype results cannot automatically be considered medical-grade.

5. "An Intelligent Garment for Long COVID-19 Real-Time Monitoring"

Authors: Marc Junior Nkengue et al.
Year: 2024

Abstract:
This research presents an intelligent garment designed for real-time monitoring of individuals experiencing long COVID-19 symptoms. The system combines several physiological parameters, particularly **heart rate, body temperature, SpO_2, and ECG signals**, and uses explainable data-fusion techniques with multiple machine-learning models to assess health status. Rather than depending on a single parameter, the approach combines physiological information to identify patterns associated with the health condition. The study reported an accuracy of approximately **97.5%** and identified SpO_2 and ECG ST-interval characteristics as particularly influential indicators in its experimental setting. The intelligent garment concept is highly relevant to a smart belt because both systems demonstrate how physiological sensors can be embedded into wearable clothing to facilitate continuous monitoring. The work also supports the proposed idea of using multiple physiological parameters together with ML instead of relying solely on fixed thresholds.

Pros:

- Uses ECG, SpO_2, temperature, and heart-rate information.
- Demonstrates multi-parameter data fusion.
- Uses machine learning for health-status assessment.
- Wearable garment provides portability and continuous monitoring.

Cons:

- Designed around a specific health condition.
- Requires extensive physiological data for reliable ML.
- Wearable sensor placement can affect signal quality.
- Results may not generalize to every population.

6. "HOT Watch: IoT-Based Wearable Health Monitoring System"

Authors: Jhansi Bharathi Madavarapu, S. Nachiyappan, S. Rajarajeswari, N. Anusha, Nirmala Venkatachalam, Rahul Charan Bose Madavarapu, A. Ahilan
Year: 2024

Abstract:
The HOT Watch research presents an IoT-enabled wearable health-monitoring device intended to provide continuous and remote monitoring of physiological parameters. The research recognizes that wearable healthcare devices can provide convenient access to health information while reducing the need for repeated clinical measurements. The proposed system considers parameters such as body temperature, respiratory rate, heart rate, blood pressure, blood glucose, SpO_2, and ECG. The work focuses on developing a cost-effective and user-friendly wearable architecture capable of collecting physiological information and transmitting it for monitoring. This study is relevant to the proposed smart belt because it demonstrates the feasibility of integrating several health sensors into a compact wearable platform. It also highlights the importance of cost, portability, usability, and continuous monitoring when developing wearable healthcare systems.

Pros:

- Supports multiple vital parameters.
- Designed for wearable operation.
- Focuses on low-cost healthcare monitoring.
- Supports IoT-based remote monitoring.

Cons:

- Integrating many sensors can increase power consumption.
- Wearable sensors may be affected by movement.
- Requires reliable wireless connectivity.
- Does not fully address personalized ML prediction in the same manner as the proposed system.

7. "Advancing Healthcare Monitoring: Integrating Machine Learning with Innovative Wearable and Wireless Systems for Comprehensive Patient Care"

Authors: IEEE Sensors Journal authors
Year: 2024

Abstract:
This research presents an advanced healthcare-monitoring architecture that combines wearable sensors, wireless communication, and machine-learning algorithms for comprehensive patient monitoring. The system focuses on physiological measurements including **SpO_2, heart rate, and body temperature**, while intelligent algorithms are used to interpret the collected information. The integration of ML provides an opportunity to move beyond simple sensor-data visualization toward automated identification of abnormal physiological patterns. Wireless connectivity enables the collected information to be accessed remotely, which is important for modern remote patient monitoring applications. The study emphasizes that wearable sensing combined with intelligent analysis can improve the accessibility and scalability of healthcare services. Its methodology supports the concept of the proposed smart belt, particularly the integration of multiple sensors with wireless communication and ML-based health-status prediction.

Pros:

- Integrates wearable sensing and machine learning.
- Supports wireless health monitoring.
- Includes important vital parameters.
- Provides a foundation for personalized healthcare.

Cons:

- Sensor accuracy can vary with user movement.
- Wireless communication depends on network availability.
- ML performance depends on representative training data.

8. "IoT-Based Remote Patient Health Monitoring System Using Raspberry Pi and Machine Learning"

Author: Buvaneshwari S
Year: 2025

Abstract:
Buvaneshwari proposed an IoT-based remote patient monitoring system using a Raspberry Pi 4 as the central processing platform. The system integrates **MAX30100 for pulse oximetry and heart rate, AD8232 for ECG, and DHT11 for temperature and humidity monitoring**. A display provides local visualization while health data is transmitted to the **ThingSpeak cloud platform**, enabling remote monitoring and storage of patient information. The system further incorporates a Random Forest machine-learning model to detect anomalies in physiological measurements and generate early warnings. Communication services are also used for alert delivery through SMS and email. This paper is extremely relevant to the proposed project because it combines almost all of the major concepts required: ECG, SpO_2, temperature, IoT, ThingSpeak, display, and machine learning. The major difference is that the proposed project uses an **ESP32-based wearable belt** and can additionally incorporate user-profile parameters from a web application to improve personalized health-status prediction.

Pros:

- Uses ECG, SpO_2, and temperature sensors.
- Uses ThingSpeak cloud.
- Uses Random Forest for anomaly detection.
- Provides remote alerts.
- Very closely related to the proposed project.

Cons:

- Raspberry Pi is larger and generally consumes more power than ESP32.
- DHT11 is less suitable for accurate body-temperature measurement than specialized medical temperature sensors.
- ML model is primarily anomaly-oriented.
- User-specific profile information is not deeply integrated into prediction.

9. "VitaLink: IoT Based Real-Time Health Monitoring System"

Authors: Pavan Patil, Priyanka Patil, Samruddhi Deshmukh, Kirti Adoni
Year: 2025

Abstract:
VitaLink is an IoT-based real-time health-monitoring system designed to acquire multiple physiological signals in a non-invasive manner. The system monitors **ECG, EMG, temperature, SpO_2, and heart rate** using biomedical sensors and employs an ESP32 microcontroller as the main communication and processing platform. ECG information is acquired using an AD8232 sensor, while additional physiological sensors provide other vital measurements. The ESP32 transmits the collected information through Wi-Fi to a mobile application, enabling remote tracking of the user's health condition. The research also incorporates signal processing and lightweight machine-learning methods to support early detection and real-time alerts. The project emphasizes cost-effectiveness, personalization, and preventive healthcare. This work has strong relevance to the proposed smart belt because it demonstrates that an ESP32 can act as the central controller for a multi-parameter wearable healthcare device while also supporting ML-based analysis.

Pros:

- Uses ESP32 for wearable healthcare monitoring.
- Monitors ECG, SpO_2, temperature, and heart rate.
- Includes lightweight ML.
- Supports wireless remote monitoring.
- Suitable for personalized healthcare.

Cons:

- EMG increases system complexity when it is not required.
- Lightweight ML may have limited predictive capability compared with larger models.
- Requires reliable sensor placement.
- Mobile-app dependence can affect accessibility.

10. "A Portable IoT-Based Health Monitoring Framework Using ESP32 for Isolated and Home-Based Patient Care"

Authors: Hemaxi Vyas, Hitasvi Shukla, Mahesh Narandas Jivani
Year: 2025

Abstract:
Vyas, Shukla, and Jivani proposed a portable IoT-based health-monitoring framework centered around the ESP32 microcontroller for home-based and isolated patient care. The system incorporates biomedical sensors to measure important parameters including **SpO_2, heart rate, and body temperature**. Real-time data acquisition is performed by the embedded platform, and the measured information is made available through a web interface and a companion application. The integrated web-server capability of ESP32 supports convenient access to health information without requiring a large computing platform. The research demonstrates the potential of ESP32 as a low-cost and portable controller for remote health monitoring. This work is particularly useful for the proposed smart belt because the belt also requires a compact microcontroller, wireless communication, local processing, and web/cloud connectivity. The proposed project extends this concept by adding ECG

acquisition, OLED display, ThingSpeak cloud storage, and machine-learning-based health-status prediction using additional user-profile parameters.

Pros:

- Uses ESP32 as the main controller.
- Low-cost and portable.
- Supports SpO_2, heart rate, and temperature.
- Provides web-based monitoring.
- Suitable for home and isolated healthcare.

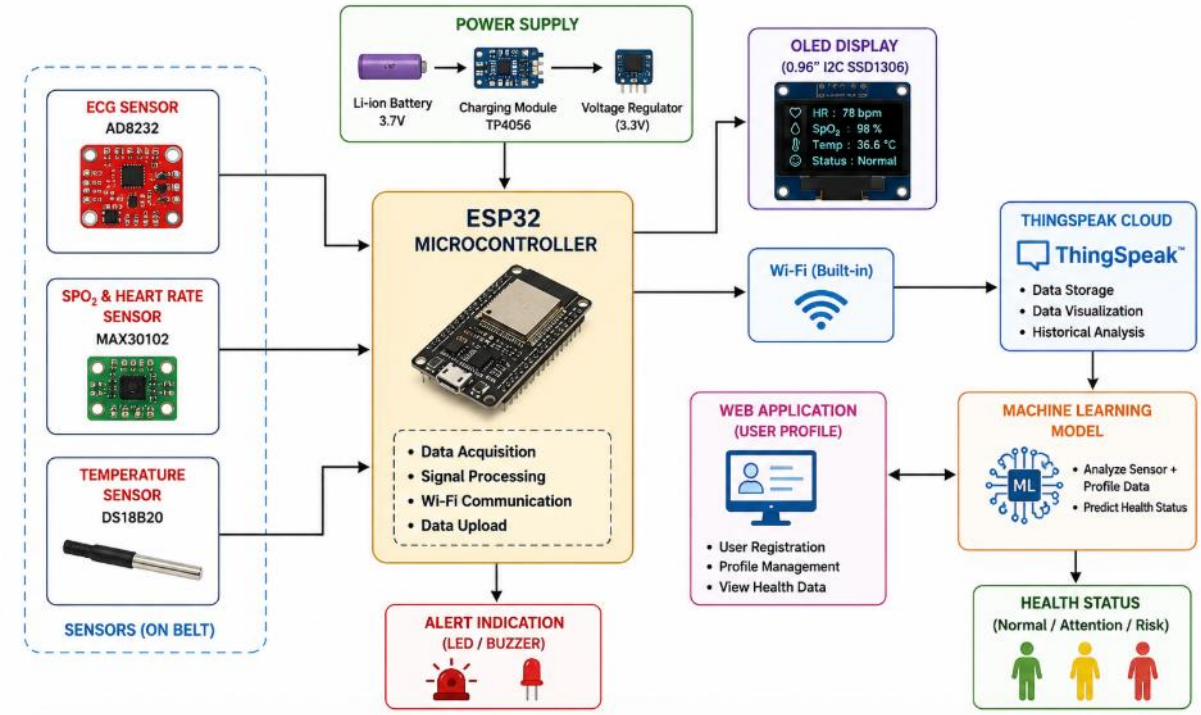
Cons:

- ECG is not included in the core reported sensor set.
- ML-based health prediction is not the primary focus.
- Web-server functionality depends on network availability.
- Additional sensors are required for comprehensive health assessment.

Objectives

1. **To develop a wearable smart health-monitoring belt** using ESP32, ECG, SpO_2, and temperature sensors for continuous acquisition of physiological parameters.
2. **To display real-time health parameters** such as ECG, SpO_2, temperature, heart rate, and predicted health status on an OLED display.
3. **To transmit and store health data on the ThingSpeak cloud** through the ESP32's built-in Wi-Fi connectivity for remote monitoring and historical analysis.
4. **To develop a web application for user-profile management** and combine profile parameters such as age, gender, height, weight, and other selected attributes with sensor data.
5. **To implement a machine-learning model** that analyzes sensor and profile parameters to predict the user's overall health status and identify potentially abnormal conditions.

Block Diagram



Detailed Methodology

1. System Design and Overall Architecture

The proposed system is developed as an **ESP32-based Smart Health Monitoring Belt** that integrates physiological sensors, local display, wireless communication, cloud storage, web-based user profiling, and machine-learning-based health-status prediction. The system is designed as a wearable platform in which the ECG, SpO_2, and temperature sensors are positioned appropriately on or near the belt so that physiological parameters can be collected continuously while the user wears the device. The ESP32 functions as the central controller and receives data from all connected sensors. It performs initial data acquisition, basic signal processing, parameter extraction, validation, and communication. The processed measurements are displayed locally using an OLED display so that the user can observe the current health parameters without depending entirely on an Internet connection. At the same time, the ESP32 uses its built-in Wi-Fi functionality to transmit the collected measurements to the ThingSpeak cloud platform. The cloud stores the measurements and provides graphical visualization of the user's health trends. A web application is additionally developed to maintain the user's profile and provide a dashboard for viewing health information. The selected profile parameters are combined with the real-time sensor data and provided to a machine-learning model, which predicts an overall health-status category.

2. Wearable Belt and Sensor Placement

The physical wearable structure consists of a comfortable belt or elastic strap designed to hold the electronic components securely while allowing the sensors to maintain appropriate

contact with the user's body. The placement of biomedical sensors is an important part of the methodology because poor contact or excessive movement can introduce noise into physiological signals. The ECG electrodes connected to the AD8232 ECG module are positioned at suitable body locations according to the selected prototype arrangement to obtain a measurable cardiac electrical signal. The MAX30102 sensor is positioned so that its optical sensing surface maintains stable contact with an appropriate measurement site, such as a finger or another suitable location depending on the mechanical design of the belt. The temperature sensor is placed in a location where it can obtain a representative body-temperature measurement. The ESP32, battery, power-management circuit, and OLED display are mounted in a protected section of the belt. The design should ensure that the user can wear the belt comfortably and that the electronic components are protected from accidental mechanical damage. Sensor placement and mechanical stability are tested before continuous data acquisition because movement-related artifacts can significantly affect ECG and optical SpO_2 measurements.

3. ECG Data Acquisition

The ECG subsystem uses an **AD8232 ECG sensor module** to acquire the electrical activity associated with cardiac activity. ECG electrodes detect the small electrical signals generated by the heart, and the AD8232 conditions and amplifies the signal so that it can be read by the ESP32. The analog ECG output is connected to an appropriate ESP32 analog input. During acquisition, the ESP32 samples the ECG signal at a suitable sampling rate selected according to the requirements of the project and the characteristics of the ECG waveform. The acquired signal is initially examined for noise, abnormal values, and signal loss. Digital filtering can be applied to reduce baseline wander, high-frequency noise, and other interference. Depending on the implementation, techniques such as moving-average filtering, low-pass filtering, high-pass filtering, or band-pass filtering can be used. After preprocessing, important ECG characteristics such as waveform pattern, heart rate, R-peaks, RR intervals, or other selected features can be extracted. These features can subsequently be used as inputs to the machine-learning model. Because ECG is highly sensitive to movement and electrode placement, the system should also monitor signal quality and avoid treating severely corrupted signals as reliable health information.

4. SpO_2 and Heart-Rate Measurement

The **MAX30102** sensor is used to obtain blood oxygen saturation and pulse-related information. The sensor uses optical measurements based on red and infrared light to estimate oxygen saturation and pulse rate. It communicates with the ESP32 through the I^2C interface. During operation, the ESP32 initializes the MAX30102 and continuously collects the optical samples produced by the sensor. The raw signal is processed to identify the pulsatile component and calculate heart-rate and SpO_2-related values. Signal averaging and basic filtering can be incorporated to reduce fluctuations caused by movement or unstable sensor contact. The resulting SpO_2 percentage and heart rate are periodically updated on the OLED display and included in the cloud data packet. If the signal quality is insufficient, the system can display an appropriate message rather than generating an unreliable health prediction. This is particularly important because optical sensors can be influenced by motion, ambient light, skin contact, and sensor positioning.

5. Body-Temperature Measurement

A suitable digital temperature sensor, such as the **DS18B20**, is incorporated into the belt for temperature monitoring. The sensor provides a digital temperature value that can be read by the ESP32 through its communication interface. During initialization, the ESP32 identifies the temperature sensor and configures it for the required measurement resolution. Temperature measurements are periodically acquired and checked for unrealistic or missing values. A small moving average or other smoothing technique can be used to reduce sudden fluctuations. The temperature value is then converted into the required unit, normally degrees Celsius, and displayed on the OLED. The same measurement is transmitted to ThingSpeak and supplied to the machine-learning pipeline. Since temperature measurements can vary depending on sensor placement and environmental conditions, the methodology should maintain consistent placement and measurement conditions during dataset collection and system testing.

6. ESP32 Data Processing and Sensor Integration

The ESP32 is responsible for coordinating all sensor operations and managing communication between the wearable hardware and external services. At startup, the microcontroller initializes the ECG input, MAX30102, temperature sensor, OLED display, Wi-Fi connection, and other required peripherals. The firmware then enters a continuous monitoring loop in which each sensor is read according to its required sampling or update rate. ECG data may be sampled continuously at a relatively higher frequency, while temperature and SpO_2 values can be updated at suitable intervals. The ESP32 performs basic preprocessing and converts raw sensor readings into meaningful parameters. It also checks whether the readings are within valid operating ranges and handles sensor disconnection or communication errors. The processed information is organized into a structured data record containing parameters such as ECG-derived features, heart rate, SpO_2, temperature, timestamp, and device/user identification information where appropriate. This common data structure allows the same measurements to be displayed locally, transmitted to the cloud, and used by the ML system.

7. OLED Display Implementation

A **0.96-inch SSD1306 OLED display** is connected to the ESP32 through the I^2C interface to provide immediate local feedback. The display is programmed to show the most important real-time health parameters in a simple and readable format. A typical screen can contain heart rate, SpO_2 percentage, temperature, ECG signal information, and the currently predicted health-status category. Because the display has limited physical size, the information can be divided into multiple screens that automatically rotate or can be selected using a button. The OLED also displays system conditions such as Wi-Fi connection status, sensor errors, or cloud-upload status. If the ML system identifies a predefined risk category, the display can show a warning message. This local feedback mechanism ensures that the user can access basic monitoring information directly from the belt without continuously opening the web application.

8. Wi-Fi Communication and ThingSpeak Cloud Integration

The ESP32's integrated Wi-Fi module is used to connect the smart belt to an available wireless network. After obtaining a network connection, the ESP32 communicates with the **ThingSpeak cloud platform** using the configured API credentials and channel information. Separate ThingSpeak fields can be assigned to parameters such as heart rate, SpO_2,

temperature, ECG-derived values, and ML-related status information. The ESP32 periodically prepares a data packet and uploads the latest measurements to the corresponding ThingSpeak fields. The cloud platform stores the information with timestamps and provides graphical visualization of the recorded measurements. Historical graphs can be used to identify gradual changes or unusual patterns in the user's physiological parameters. If the Internet connection is temporarily unavailable, the firmware can detect the communication failure and continue local monitoring, while attempting to reconnect to the network periodically. This makes the system more robust than a design that stops functioning completely when cloud connectivity is interrupted.

9. Web Application and User Profile Development

A web application is developed to provide a personalized interface for the health-monitoring system. The application includes user registration, login, profile management, health-data visualization, and health-status display functions. The user profile contains selected parameters such as **age, gender, height, weight, and other project-defined attributes** that can contribute to personalized ML analysis. The profile information is stored in a database and associated with the corresponding user's monitoring data. The web dashboard can display the latest measurements obtained from the cloud along with historical graphs and the ML-predicted health status. The application can also provide a summary of the user's recent readings and highlight measurements that differ significantly from the user's normal pattern. Appropriate authentication and access controls should be implemented so that health information is available only to authorized users. The web application therefore acts as the interface between the wearable monitoring device, cloud data, user information, and machine-learning prediction system.

10. Dataset Preparation for Machine Learning

The machine-learning component requires a suitable dataset containing physiological measurements and corresponding health-status labels. The dataset can be obtained from appropriate publicly available research datasets or collected under a properly designed and ethically appropriate data-collection procedure. Each record should contain relevant features such as ECG-derived parameters, heart rate, SpO_2, body temperature, and selected user-profile parameters. The target variable represents the health-status class defined for the project, for example **Normal, Attention Required, and Risk**. Before training, the dataset is checked for missing values, duplicate records, invalid measurements, and inconsistent labels. Missing numerical values can be handled using suitable imputation methods, while obviously erroneous sensor values should be removed or corrected according to predefined rules. Numerical features may be normalized or standardized where required by the selected ML algorithm. Categorical variables such as gender can be converted into a machine-readable representation. The dataset is then divided into training and testing subsets, with an optional validation subset for model tuning.

11. Feature Engineering and ML Model Training

Feature engineering is performed to identify the most useful information for predicting health status. Direct features may include temperature, SpO_2, heart rate, age, height, and weight, while ECG can be represented using extracted characteristics such as heart rate, RR interval statistics, waveform-related features, or other validated ECG features. Derived features such as body-mass index can also be calculated from height and weight if appropriate for the

project. Several supervised machine-learning algorithms can then be evaluated, including **Random Forest, Logistic Regression, K-Nearest Neighbors, Decision Tree, or Support Vector Machine**. Random Forest is particularly suitable as a baseline because it can handle nonlinear relationships and mixed feature types while providing feature-importance information. The models are trained using the training dataset and evaluated using the testing dataset. Performance metrics such as accuracy, precision, recall, F1-score, and confusion matrix are calculated. Cross-validation can additionally be used to reduce the possibility that the model's performance depends on a particular train-test split. The best-performing model is selected based not only on accuracy but also on its ability to identify potentially important risk classes reliably.

12. Integration of User Profile with Sensor Data

A key feature of the proposed methodology is the integration of **real-time physiological measurements with information obtained from the web application's user profile**. When a user is registered, the selected profile parameters are stored in the database. During health-status prediction, the latest sensor measurements are retrieved from the ESP32/ThingSpeak pipeline and combined with the corresponding profile information. For example, a prediction record may contain age, gender, height, weight, heart rate, SpO_2, temperature, and ECG-derived features. The same preprocessing operations used during model training must be applied to incoming data before prediction. This ensures that the model receives data in the same format and scale as the training dataset. The combined feature vector is then supplied to the trained ML model, which generates a health-status prediction. This personalized approach allows the system to consider both current physiological measurements and selected characteristics of the individual instead of treating every user identically.

13. Real-Time Health-Status Prediction

After the ML model has been trained and saved, it is integrated with the web application or an appropriate prediction service. Whenever a new set of valid sensor measurements becomes available, the system constructs a prediction input using the sensor parameters and user-profile information. The input is passed through the same preprocessing pipeline used during training and then supplied to the trained model. The model generates a predefined health-status class. For example, the output can be represented as **Normal**, **Attention Required**, or **Risk**. The prediction is returned to the web dashboard and can also be uploaded to ThingSpeak for historical tracking. The OLED display can show a simplified status to the wearer. If the project includes an LED or buzzer, a warning can be activated for the predefined risk category. The system should avoid treating a single noisy measurement as a definitive health condition; instead, multiple consecutive measurements or a confidence-based decision mechanism can be used to reduce false alarms.

14. Cloud-Based Historical Analysis

The ThingSpeak cloud platform serves as the central storage and visualization layer for the sensor measurements. Each uploaded record is associated with a timestamp, allowing the system to construct historical graphs for ECG-derived measurements, heart rate, SpO_2, and temperature. These graphs can be analyzed to identify trends such as gradual temperature changes, repeated fluctuations in SpO_2, or variations in heart rate. The historical information can also support future ML development by providing additional time-series data, subject to appropriate data governance and privacy controls. The web application can retrieve the cloud

information and present it in a more user-friendly dashboard. The combination of real-time and historical information allows the proposed system to move beyond simple instantaneous monitoring toward continuous health observation.

15. Alert and Abnormal-Condition Handling

An alert mechanism can be incorporated to notify the user when sensor readings or ML predictions indicate a potentially abnormal condition. The system can use a combination of sensor-validity checks, predefined safety thresholds, and ML output rather than depending exclusively on one mechanism. For example, if the SpO_2 measurement remains unusually low over multiple valid readings, or if the temperature and other parameters form a pattern associated with the selected risk class, the system can generate an alert. The OLED can display a warning, while an LED or buzzer can provide a local indication. The web application can additionally record the event and, in an expanded implementation, send a notification to an authorized caregiver. Alert logic should be carefully designed to minimize false alarms caused by temporary sensor errors, movement artifacts, or poor contact.

16. System Testing and Validation

The complete system is tested in multiple stages to verify hardware, software, communication, and ML performance. Initially, each sensor is tested independently to confirm that it produces stable and meaningful measurements. The ECG subsystem is checked for waveform quality, the MAX30102 is evaluated for stable pulse and SpO_2 readings, and the temperature sensor is compared against an appropriate reference measurement. The OLED is tested for correct display updates, while Wi-Fi and ThingSpeak communication are tested under different network conditions. After individual testing, all sensors are operated simultaneously to evaluate whether there are electrical or software conflicts. The web application is then tested for user registration, profile storage, data retrieval, dashboard visualization, and prediction functionality. Finally, the ML model is evaluated using an independent test dataset and appropriate classification metrics. The complete belt is tested under controlled movement and normal usage conditions to identify signal-quality problems and communication failures.

17. Complete Operational Workflow

During normal operation, the user first wears the smart belt and switches on the device. The ESP32 initializes all sensors and establishes the Wi-Fi connection. The ECG sensor continuously acquires cardiac electrical signals, the MAX30102 obtains SpO_2 and heart-rate information, and the temperature sensor measures body temperature. The ESP32 preprocesses the acquired data and displays the latest valid measurements on the OLED. At defined intervals, the processed values are transmitted to ThingSpeak through Wi-Fi. The web application retrieves the corresponding measurements and associates them with the user's stored profile. The combined sensor and profile data are passed through the ML preprocessing and prediction pipeline. The trained model produces the predefined health-status classification, which is displayed on the web application and, where implemented, on the OLED. If an abnormal pattern is identified according to the project's decision logic, an alert can be generated. The complete process is repeated continuously, allowing the system to provide real-time monitoring together with cloud-based historical analysis and personalized ML-based health-status estimation.

Hardware Requirements

- **ESP32 Development Board** – Main controller for sensor acquisition, processing, Wi-Fi communication, and ML-related data handling.
- **AD8232 ECG Sensor Module** – For acquiring the electrical activity of the heart.
- **MAX30102 Sensor** – For measuring SpO_2 and heart rate/pulse rate.
- **Body Temperature Sensor** – DS18B20 or another suitable contact/non-contact temperature sensor for measuring body temperature.
- **OLED Display (0.96-inch I^2C, SSD1306)** – For displaying real-time sensor readings and health-status information.
- **Li-ion/Li-Po Rechargeable Battery** – Portable power supply for the wearable belt.
- **Battery Charging Module** – For safe charging of the rechargeable battery.
- **Voltage Regulator/Power Management Module** – To provide appropriate voltage levels to the ESP32 and sensors.
- **Smart Belt/Elastic Wearable Strap** – To securely mount the sensors and electronics on the user's body.
- **ECG Electrodes and Connecting Wires** – For obtaining ECG signals from the user.
- **Jumper Wires/PCB/Connectors** – For electrical connections between the ESP32 and sensors.
- **Buzzer/LED (Optional)** – For providing an alert when abnormal health conditions are detected.
- **Enclosure/Protective Casing** – To protect the electronics and improve wearable safety.

Software Requirements

- **Arduino IDE** – For programming and uploading firmware to the ESP32.
- **Embedded C/C++** – Programming language used for ESP32 firmware development.
- **ESP32 Board Package** – Required to compile and upload programs to the ESP32.
- **Adafruit SSD1306 / Adafruit GFX Libraries** – For controlling the OLED display.
- **MAX3010x Sensor Library** – For interfacing the MAX30102 sensor and obtaining SpO_2/heart-rate data.
- **AD8232 ECG Interface Code/Signal Processing** – For acquiring and preprocessing ECG signals.
- **OneWire and DallasTemperature Libraries** – Required when using a DS18B20 temperature sensor.
- **Wi-Fi Libraries for ESP32** – For connecting the device to the Internet.
- **ThingSpeak Cloud Platform** – For uploading, storing, visualizing, and monitoring sensor data remotely.
- **Python** – For dataset preparation, machine-learning model development, evaluation, and prediction.
- **Pandas and NumPy** – For data processing and numerical operations.
- **Scikit-learn** – For implementing models such as Random Forest, Logistic Regression, KNN, or Decision Tree.
- **Flask/PHP** – For developing the web application and user-profile management system.
- **HTML, CSS, JavaScript, Bootstrap** – For designing the web-based health-monitoring interface.

Conclusion

The proposed **ESP32-Based Smart Health Monitoring Belt** provides an integrated and portable solution for continuous monitoring of important physiological parameters using modern IoT and machine-learning technologies. The system combines an **ESP32 microcontroller, ECG sensor, SpO_2 sensor, temperature sensor, and OLED display** to acquire and present real-time health information in a convenient wearable form. The ESP32 acts as the central processing unit, collecting data from the sensors, performing basic signal processing, and displaying parameters such as ECG information, oxygen saturation, heart rate, and body temperature on the OLED display. Through the built-in Wi-Fi capability of the ESP32, the collected health information can be transmitted to the **ThingSpeak cloud platform**, where it can be stored and visualized for continuous monitoring and historical analysis. This cloud connectivity makes the system useful not only for the wearer but also for remote monitoring applications where authorized caregivers or healthcare personnel may need access to health trends.

An important feature of the proposed system is the integration of a **web application and machine-learning model** with the wearable device. The web application can maintain a user profile containing parameters such as age, gender, height, weight, and other selected characteristics that may influence the interpretation of physiological measurements. Instead of analyzing ECG, SpO_2, and temperature independently, these parameters can be combined with suitable profile information and provided as input to a trained machine-learning model. The model can identify patterns in the collected data and classify the user's condition into predefined categories such as **Normal, Attention Required, or Risk/Critical**, depending on the training dataset and project requirements. This approach can provide more personalized health-status estimation than a simple fixed-threshold monitoring system. Overall, the project demonstrates the effective integration of **embedded systems, biomedical sensors, wearable technology, IoT cloud computing, web applications, and machine learning** into a single healthcare-monitoring platform. The system can serve as a useful academic prototype for continuous health observation and early indication of potentially abnormal physiological patterns, while actual medical diagnosis would require clinically validated sensors, datasets, testing, and regulatory approval.

Future Scope

The future scope of the smart health-monitoring belt is extensive because the basic ESP32-based architecture can be expanded with additional sensors, advanced machine-learning algorithms, improved communication technologies, and more sophisticated healthcare applications. One important improvement would be the addition of sensors for **blood pressure, respiratory rate, body movement, fall detection, and activity monitoring**. An MPU6050 or similar inertial sensor could be incorporated to detect falls, posture changes, physical activity, and abnormal movement patterns. A respiratory sensor could provide breathing-rate information, while additional biomedical sensors could enable more comprehensive physiological monitoring. GPS could also be integrated for location tracking during emergency situations, allowing the system to transmit the user's approximate location to authorized contacts when a serious abnormal event is detected.

Another major area of future development is the improvement of the **machine-learning and artificial-intelligence component**. The initial system can use models such as Random

Forest, Logistic Regression, KNN, or Decision Tree for health-status classification. Future versions could evaluate more advanced approaches such as **LSTM, CNN, hybrid CNN-LSTM, autoencoders, or lightweight edge-AI models** for analyzing ECG waveforms and time-series physiological data. Rather than predicting health status from individual readings, the model could learn long-term patterns from historical data collected through ThingSpeak and the web application. Personalized models could be developed by considering individual baseline measurements and profile information. Explainable AI techniques could also be incorporated so that the system can indicate which parameters contributed most to a particular prediction, making the output easier to understand and potentially more useful for healthcare professionals.

The **web application and cloud platform** can also be significantly enhanced. The current system can be extended into a complete healthcare dashboard containing real-time sensor graphs, historical trends, health-status reports, user profiles, and configurable alerts. Authorized caregivers could receive notifications through email, SMS, or mobile applications when predefined abnormal patterns are detected. A mobile application could be developed to allow users to monitor their health information from smartphones. Secure authentication and role-based access could be implemented to ensure that only authorized users can access sensitive health information. Cloud databases could also be used to maintain long-term health records, while appropriate encryption and privacy mechanisms would be necessary to protect the collected data.

The physical design of the belt can also be improved to make it more comfortable and reliable for long-term use. Future versions can use **flexible PCB technology, conductive fabric electrodes, smaller sensors, rechargeable battery management, and low-power communication techniques** to reduce the size and weight of the device. Sensor placement can be optimized to minimize motion artifacts, particularly in ECG measurements. The system could also incorporate adaptive filtering and advanced signal-processing techniques to improve ECG and SpO_2 signal quality during movement. Finally, the most important future step would be **clinical validation** using appropriately collected and ethically governed datasets from diverse populations. The ML model should be evaluated for accuracy, sensitivity, specificity, false-alarm rate, robustness, and generalization before being considered for real healthcare applications. With these improvements, the proposed smart belt could evolve from an academic IoT prototype into a more comprehensive **personalized wearable health-monitoring and early-warning platform**.

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